FINAL PROJECT

ENGY 644

Two-Stage Stochastic Renewable Procurement for a Microgrid in College Station

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1 EXECUTIVE SUMMARY

The main objective of this project is to determine the best optimal solution for day-ahead renewable energy procurement for a College Station microgrid under uncertain weather conditions. Both AMPL and Python were utilized to develop models to determine the stochastic, expected value, and wait-and-see objectives in order to determine the best optimization of the objective to minimize total expected system cost by determining the commitment schedules of energy from solar and wind sources. The results across each model show that the optimal strategy differentiates between solar and wind resources based on time-of-day availability. Solar generation dominates during daylight hours while wind generation is primarily used in the evening and at night. Scenario-based results show that system costs are sensitive to weather conditions. Comparison of the models through EVPI and VSS analysis suggest that a model that takes uncertainty into account performs better for real-world scenarios. Further sensitivity testing through variable grid purchase costs during hours with peak demand indicate that the system faces constraints by renewable generational capacity. Both AMPL and Python produced nearly identical models, reinforcing the reliability of the findings and conclusions in this report.

2 INTRODUCTION

Microgrids are a popular method to improve energy resilience, reduce emissions, and manage electricity costs for small-scale operations. In regions such as College Station, Texas, renewable energy resources offer significant potential for reducing reliance on grid electricity. However, resources like solar and wind have innate uncertainty due to weather variability, which can make day-ahead procurement decisions challenging.

Solar generation is highly dependent on time of day and cloud cover, typically peaking during midday hours and dropping to zero overnight. Wind generation, while often available during nighttime and early morning periods, is less predictable and can vary significantly across weather conditions. At the same time, electricity demand follows its own daily pattern, often peaking in the late afternoon and evening in response to the day's heat and the setting of the sun.

Traditional deterministic optimization approaches assume that future conditions are known or represented by average values. While this simplifies the problem, it fails to capture variability in renewable generation and may produce solutions that perform poorly under real-world conditions. To address this, this project uses a two-stage stochastic linear programming framework, where first-stage decisions are made before uncertainty is realized and second-stage adjustments occur after scenarios are observed. This approach captures uncertainty and supports more robust decision-making.

3 METHODS

3.1 Mathematical Formulation

The microgrid procurement problem is formulated as a two-stage stochastic linear program. The model is defined over a set of hourly time periods and a set of discrete weather scenarios representing different combinations of solar and wind conditions.

Sets

- T: Set of hourly time periods, $t = 1, \ldots, 24$
- Ω: Set of scenarios, indexed by ω

Parameters

- $D_{t,\omega}$: Electricity demand in hour t under scenario ω
- $q^S_{t,\omega}$: Solar availability factor in hour t, scenario ω
- $q^W_{t,\omega}$: Wind availability factor in hour t, scenario ω
- c^S_t, c^W_t: Contract costs for solar and wind
- p^G_t: Grid purchase price
- p^E_t: Export price
- U^S_t, U^W_t: Maximum solar and wind contract limits
- π_ω: Probability of scenario ω

Decision Variables

First-stage (here-and-now):

- x^S_t: Solar energy contracted in hour t
- x^W_t: Wind energy contracted in hour t

Second-stage (recourse):

- $g_{t,\omega}$: Grid electricity purchased
- $e_{t,\omega}$: Electricity exported

Objective Function

The objective is to minimize total expected cost:

$$\min \sum_{t \in T} \left(c^S_t x^S_t + c^W_t x^W_t \right) + \sum_{\omega \in \Omega} \pi_\omega \sum_{t \in T} \left(p^G_t g_{t,\omega} - p^E_t e_{t,\omega} \right)$$

Constraints

Energy Balance:

$$q^S_{t,\omega} x^S_t + q^W_{t,\omega} x^W_t + g_{t,\omega} - e_{t,\omega} = D_{t,\omega}$$

Capacity Limits:

$$x^S_t \leq U^S_t, \quad x^W_t \leq U^W_t$$

The first-stage decisions for committed solar and wind blocks are made before the realization of uncertainty and must therefore be identical across all scenarios. This ensures that decisions are made without knowledge of future weather conditions, reflecting real-world operational constraints and satisfying the nonanticipativity constraint.

These models were ultimately used to calculate the Expected Value of Perfect Information (EVPI) by subtracting the optimal value of the stochastic model by the optimal value of the deterministic “wait-and-see" model, calculated as the sum of the probability times the optimal value for each scenario. The Value of Stochastic Solution (VSS) was also calculated from these models by subtracting the optimal value of the stochastic model from the optimal solution from the expected value solution (EEV). These two calculations are shown below:

$$EVPI = z^{SP} - z^{WS}, \quad \text{with } z^{WS} = \Sigma a_\omega z_\omega^* \qquad VSS = z^{EEV} - z^{SP}$$

The objective values for each model, the EVPI and VSS for both languages were then compared in order to ensure the values obtained from both languages accurately reflect the models.

Finally, to evaluate the robustness of the stochastic solution, a sensitivity analysis was conducted by increasing grid purchase prices by 20% during peak demand during hours 16-20. This represents a higher-cost operating environment during late afternoon and evening periods, when electricity demand is typically highest and grid prices are most volatile.

3.2 Computational Implementation

Both Python and AMPL were implemented separately in this study to compare consistency, solver performance, and modeling approaches.

3.2.1 Python

The stochastic optimization model was implemented in Python using the Pyomo optimization library. The model was solved using the GLPK linear programming solver within a Google Colab environment. Input data were provided as CSV files containing hourly economic parameters, demand values, and scenario-based solar and wind capacity factors. The stochastic structure was represented using eight discrete weather scenarios and their associated probabilities. The deterministic equivalent of the stochastic program was implemented by explicitly expanding the model across all hours and scenarios, allowing the solver to optimize expected cost directly. All model parameters, including demand, renewable availability, prices, and capacity limits, were converted into dictionary formats compatible with Pyomo. Finally, key visualizations were created to represent the final results

3.2.2 AMPL

The AMPL models were coded in AMPL IDE. Separate .mod files were created for each of the models (2-stage stochastic, expected value, and deterministic). One data file containing all given parameters and given data was made for all of the models, outlined using the intrinsic language for AMPL. One run file was made for each model, with separate blocks for running each model with a reset in between. This also allowed for the first-stage decision variables to be the same across all scenarios, since they were called using the same variable term each time. The calculations for

EVPI and VSS were conducted at the end of the run file, with some values being recorded from previous runs of the model. A text file was also made to outline the structure of the documentation. The visualizations were created in Microsoft Excel, due to the lack of a visualization package within the AMPL environment.

4 RESULTS

4.1 Stochastic Model

The results for the stochastic model as determined by the respective models are shown in the figures below. In both figures, solar is represented in blue and wind is represented in orange.

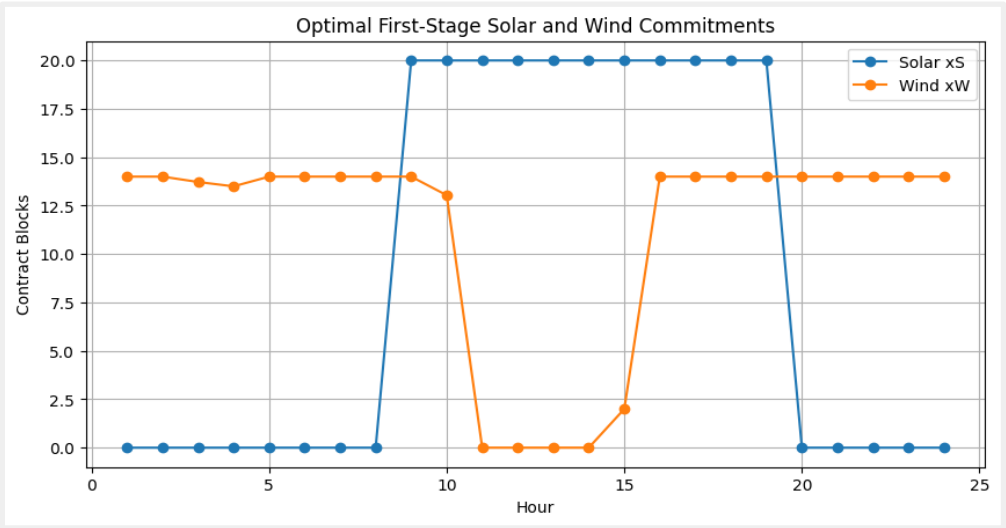

Figure 1. Python Stochastic First-Stage Commitment Schedules for Solar and Wind

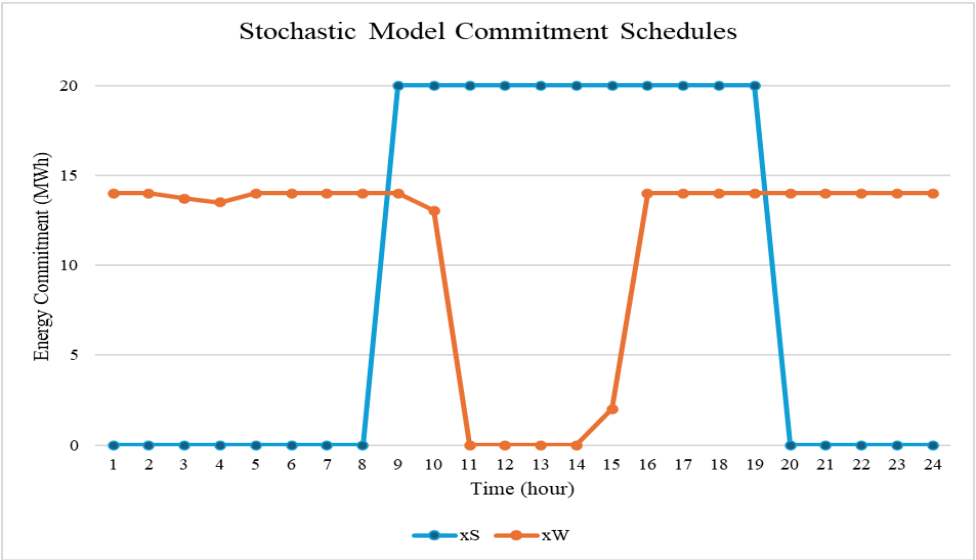

Figure 2. AMPL Stochastic First–Stage Commitments for Solar and Wind

Additionally, the second-stage recourse costs from grid purchases and exports were formatted into a table in the Python model (Appendix Table A) and graphically from the AMPL model (Appendix Figures A and B). The total cost per scenario for the stochastic model was graphed in Figure C in the appendix.

4.2 Expected Value Model

The commitment schedules from the expected value model in the Python code were graphed in comparison to the results from the stochastic model, while in AMPL the schedules were graphed separately. The results are shown in figures 3 and 4 below, with EV solar commitments in orange and the EV wind commitments in red.

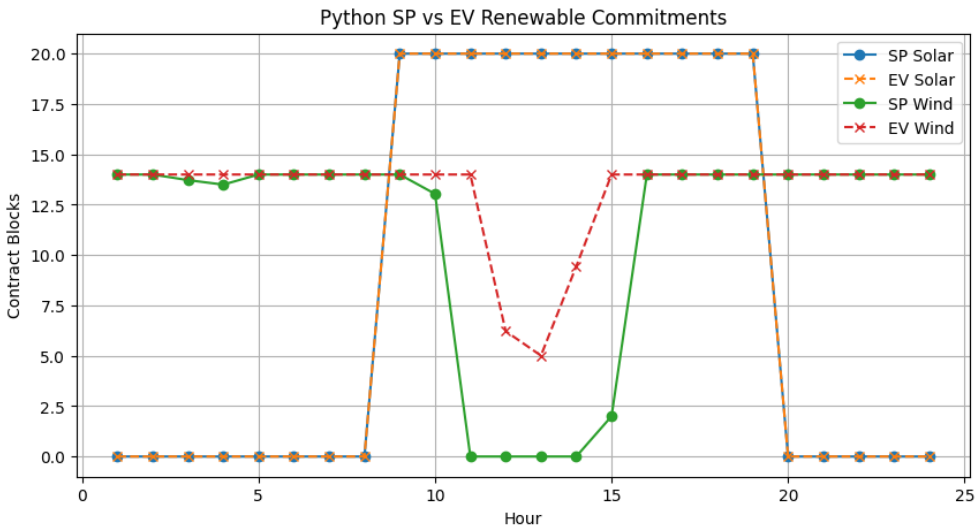

Figure 3. Python EV Commitments for Solar and Wind

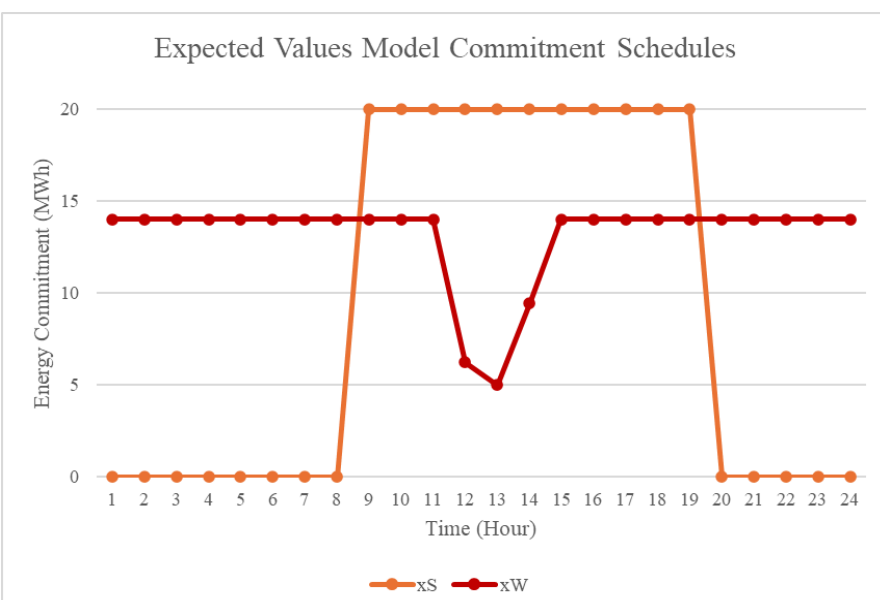

Figure 4. AMPL EV Commitments for Solar and Wind

4.3 Wait-and-See Model

The results from the deterministic wait-and-see models for both models were obtained from their respective models and inputted into the combined table 1, shown below.

Scenario	Probability	Optimal Cost, AMPL ($)	Weighted Optimal Cost, AMPL ($)	Optimal Cost, Python ($)	Weighted Optimal Cost, Python ($)
Hot Clear	0.16	11,371.95	1,819.51	11,371.95	1,819.51
Hot Hazy	0.14	13,793.75	1,931.13	13,793.75	1,931.13
Hot Cloudy	0.12	14,446.52	1,733.58	14,446.52	1,733.58
Storm PM	0.10	13,203.28	1,320.33	13,203.28	1,320.33
Clear Breezy	0.14	8,336.04	1,167.05	8,336.04	1,167.05
Mild Balanced	0.14	8,753.17	1,225.44	8,753.17	1,225.44
Overcast Windy	0.10	10,833.11	1,083.31	10,833.12	1,083.31
Cloudy Calm	0.10	14,960.13	1,496.01	14,960.13	1,496.01
Total Cost	**-**	-	**11,776.36**	**-**	**11,776.36**

Table 1 - Wait-and-See Model Results Comparison

4.4 Sensitivity Analysis

The sensitivity of the models to changes in grid price during peak hours was recorded as seen in Table 2. Figure D in the appendix shows a comparison of changes due to sensitivity.

Hour	Old Grid Price ($)	New Grid Price ($)
16	98.0	117.6
17	108.0	129.6
18	116.0	139.2
19	112.0	134.4

Table 2 - Updated hourly grid costs for sensitivity analysis from the AMPL model.

The new objective value was determined to be $12,818.31, which is ~$795.79 more than the original stochastic objective solution.

4.5 - Summary of Results

Language	AMPL	Python
Stochastic Optimal Total Cost	$12,022.50	$12,022.52
Expected Value Optimal Total Cost	$11,518.34	$11,518.34
Deterministic Optimal Total Cost	$11,776.36	$11,776.36
EVPI	246.14	246.16
VSS	108.69	108.67

Table 3 - Model Results Comparison

5 ANALYSIS AND DISCUSSION

5.1 Computational Model Validation

To verify the correctness and robustness of the optimization models, both Python (Pyomo with the GLPK solver) and AMPL (Gurobi solver) were used. The objective values obtained from the two modeling environments were highly consistent, with negligible differences likely due to rounding differences between the solvers. The calculated values for EVPI and VSS were also similar, with slight decimal variations. Comparison of exacted values obtained from each model can be seen in Table 3. From the similar findings in this project, it can be assumed that the solution is not dependent on a specific modeling framework. From an engineering perspective, this cross-platform consistency increases confidence in both the numerical results and the underlying decision insights.

5.2 Solar and Wind Commitments Across Scenarios

The optimal first-stage commitment schedules determined from the stochastic models are shown in Figures 1 and 2. Solar commitments are concentrated entirely during daylight hours, specifically from hour 9 through hour 19. The solar commitment during these hours is at the maximum commitment of 20 blocks. Outside of this window, solar commitments are zero while the sun is down for the night. During these hours, the commitments switch to wind. During nighttime hours 1 through 8, wind is utilized at or near its maximum capacity of 14 blocks, serving as the primary renewable resource while solar is unavailable. As the sun rises and solar generation becomes available, the two energy sources mix. Wind commitments reach zero during peak solar generation for hours 11 through 14. These patterns align well with expected engineering behavior for College Station summer conditions; solar generation peaks during midday due to strong sunlight, while

wind generation tends to be more available during nighttime and early morning hours. Thus, the model effectively allocates resources in a way that maximizes renewable utilization while simultaneously minimizing cost. This also allows for the model to maintain flexibility in cases of uncertainty, both due to the availability of solar at higher commitment capacity during the day and through the continuous availability of wind.

Different weather scenarios were shown to have significant effects on the total cost, as shown in Figures A and C in the appendix. The most expensive scenario determined was the Cloudy Calm scenario at $15,375.80, likely characterized by limited solar generation from lack of sun and limited wind generation due to decreased availability. During peak hours in this scenario, particularly in the late afternoon and evening, grid purchases increase substantially in order to close the gap between electricity generation from the two renewables and the demand of the hour. In contrast, the lowest-cost scenario was the Clear Breezy weather conditions, in which there was ample sunshine and wind availability to produce both solar and wind for electricity, thus decreasing the need for grid purchases and even increasing electricity exports, as seen in Figure B in the appendix. The weighted contributions of each scenario, determined by the deterministic model and shown in Table 1, also provide proof of this case, with the scenarios of higher solar and/or higher wind availability contributing a lower total cost to the summed total cost solution. However, these costs are also impacted by their likeliness of occurring, numerically determined through the probability. Thus, it is still important to maintain the option of grid purchase in case the renewable sources are unable to meet demand, and to build in flexibility for uncertainty overall.

5.3 Uncertainty Analysis

The expected value solution (EEV) was calculated in order to evaluate the importance of modeling uncertainty. The EV model replaces all scenario-dependent parameters with their probability-weighted averages, effectively assuming weather conditions will act according to their probabilities. As a result, the EV model ignores variability and produces a lower objective value of approximately $11,518.34 compared to the stochastic solution value of ~$12,023, as determined in both the AMPL and Python models.

Upon further analysis of the stochastic model versus the deterministic EV model, the value of the stochastic model (VSS) was calculated in each model in order to determine the effect that uncertainty has on total cost. The VSS quantifies the difference between the EEV and the stochastic optimal solution. For these models, the VSS, calculated to be positive and slightly over 108 in both language models, indicates that incorporating uncertainty leads to a measurable improvement in expected cost. Essentially, the stochastic model has a value equal to $108 that can be saved in the total cost because of the incorporation of uncertainty. However, this is a moderate value because the total costs are $11,000-12,000+, meaning $108 is less than 1% of the total cost.

The stochastic solution was also compared to the wait-and-see model (WS) to determine the maximum benefit of having perfect forecasts. The EVPI, calculated to be slightly over 246 in both

language models, is positive but relatively limited. This indicates that, while better forecasts would reduce costs, the current stochastic solution already captures much of the available value by hedging, or mixing both solar and wind, across the different weather scenarios. Therefore, even having a perfect forecast would not influence the optimal solution that much because of the already adequate performance of the stochastic model.

5.4 Sensitivity Analysis

The results of the sensitivity analyses from both models show that the total system cost increases relative to the base case, reflecting the higher cost of grid purchases during peak hours. However, the first-stage renewable commitment decisions remain largely unchanged compared to the original models. This outcome suggests that the original stochastic solution was already operating at or near the available renewable capacity limits during peak hours. As a result, the system does not have the flexibility to increase renewable procurement in response to higher grid prices and is instead absorbed through higher recourse costs.

5.5 Practical Applications for Microgrid in College Station

The results from these models provide several important insights that can be considered for microgrid operation in College Station. First, it is important to note that solar should always be fully utilized during peak daytime hours because availability is high and therefore cost is low, evident in the solar and wind commitment graphs above. In a greater sense, this can also be applied to other energy sources, such as natural gas and hydroelectricity, to balance energy production by using what is most available at the time. The balance, however, is also important, because it diversifies an energy portfolio such that, in cases where availability for one energy source is not high enough, another source can close the gap. This is evident in Table 1, in which scenarios with low sun and wind availability proved to be more expensive due to a need to purchase electricity from the grid, shown in Figure A in the appendix. Also, during nighttime when solar was unavailable, wind was able to provide electricity and thus balance the energy demand.

The comparison of each model also shows the benefits of creating a stochastic model over a model that does not account for uncertainty. Though the final optimal solutions shown in Table 3 make it seem like the stochastic model is more expensive, the VSS and EVPI analyses indicate that inflexibility to uncertainty has a negative impact on the objectives. This is reflected in Table 1, in which the weather scenario can have a major impact on the total cost. The stochastic model is the best model for the real-world, since weather can be unpredictable.

6 CONCLUSION

This project demonstrates the effectiveness of two-stage stochastic optimization for microgrid renewable energy procurement under uncertainty compared to deterministic models. By incorporating multiple weather scenarios and their associated probabilities, the stochastic model

captures the variability of solar and wind generation and produces decisions that are robust across a range of weather scenarios. The models indicated that solar should be the dominant energy source during the day, while wind dominates at night, but also with the flexibility for solar and wind to operate simultaneously during the day at peak electricity usage hours. Although deterministic models provide a useful baseline, they underestimate costs and inadequately model uncertainty. The stochastic approach improves decision quality and reduces overall expected cost in a way that best reflects real-life weather scenarios. Overall, it can be determined that effective microgrid operation requires both diversification of power sources and flexibility in planning, and that stochastic optimization provides the best practical framework for achieving these objectives.

APPENDIX

Scenario	Probability	First Stage Cost	Recourse Cost	Total Cost	Weighted Total Cost
Hot Clear	0.16	4422.616375	6972.580401	11395.196776	1823.231484
Hot Hazy	0.14	4422.616375	9583.078821	14005.695196	1960.797327
Hot Cloudy	0.12	4422.616375	10278.226057	14700.842432	1764.101092
Storm PM		4422.616375	9182.771866	13605.388240	1360.538824
Clear Breezy	0.14	4422.616375	4066.134000	8488.750375	1188.425052
Mild Balanced	0.14	4422.616375	4481.668241	8904.284616	1246.599846
Overcast Windy	0.10	4422.616375	6989.866458	11412.482832	1141.248283
Cloudy Calm	0.10	4422.616375	10953.188561	15375.804936	1537.580494

Table A - Recourse Results from 2nd-stage Stochastic Model

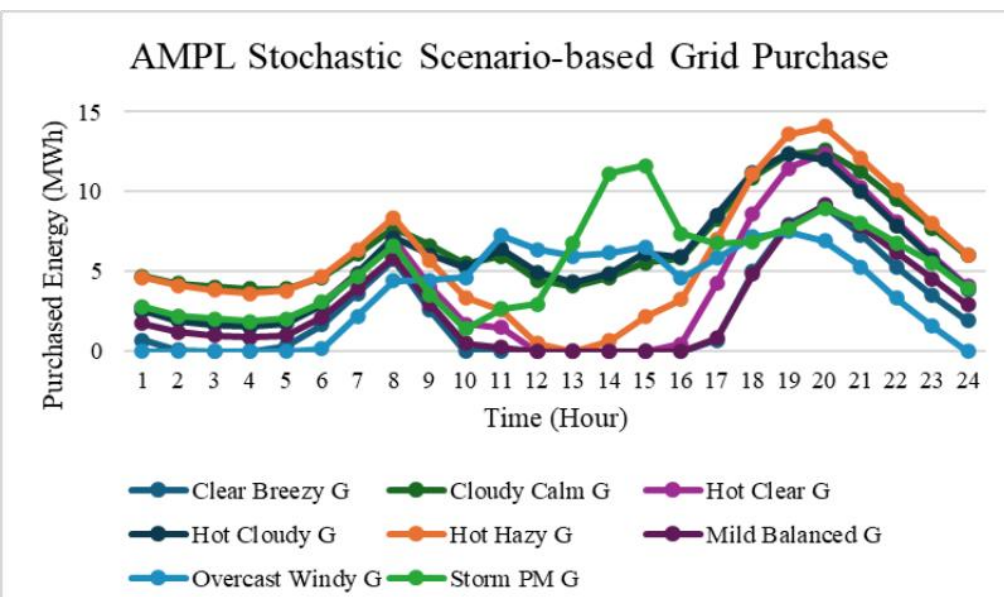

Figure A - Stochastic Scenario-based Grid Purchase from the AMPL Model

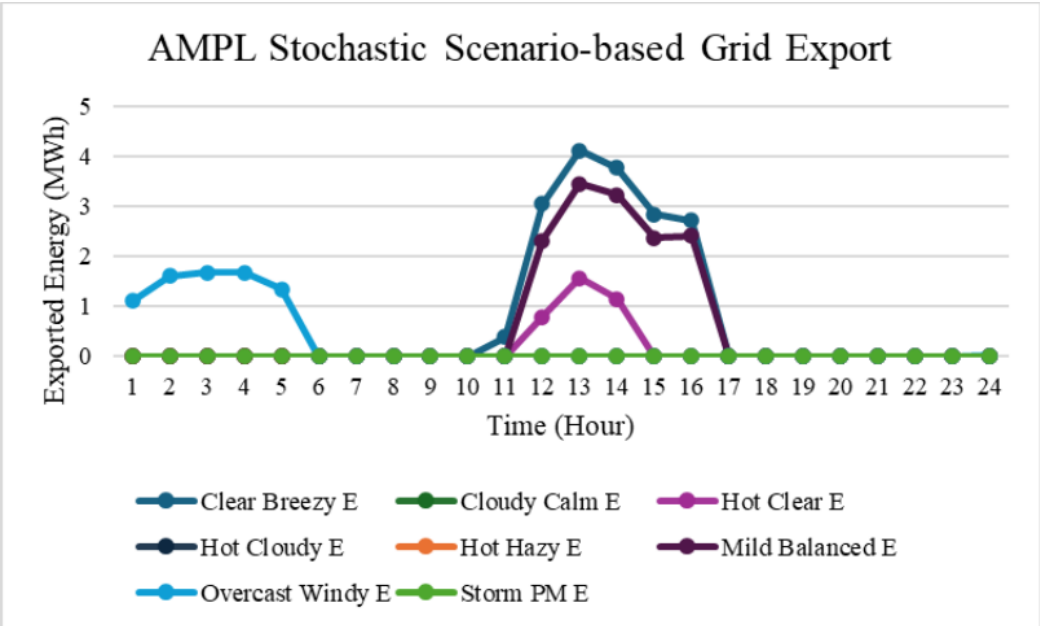

Figure B - Stochastic Scenario-based Grid Export from the AMPL Model

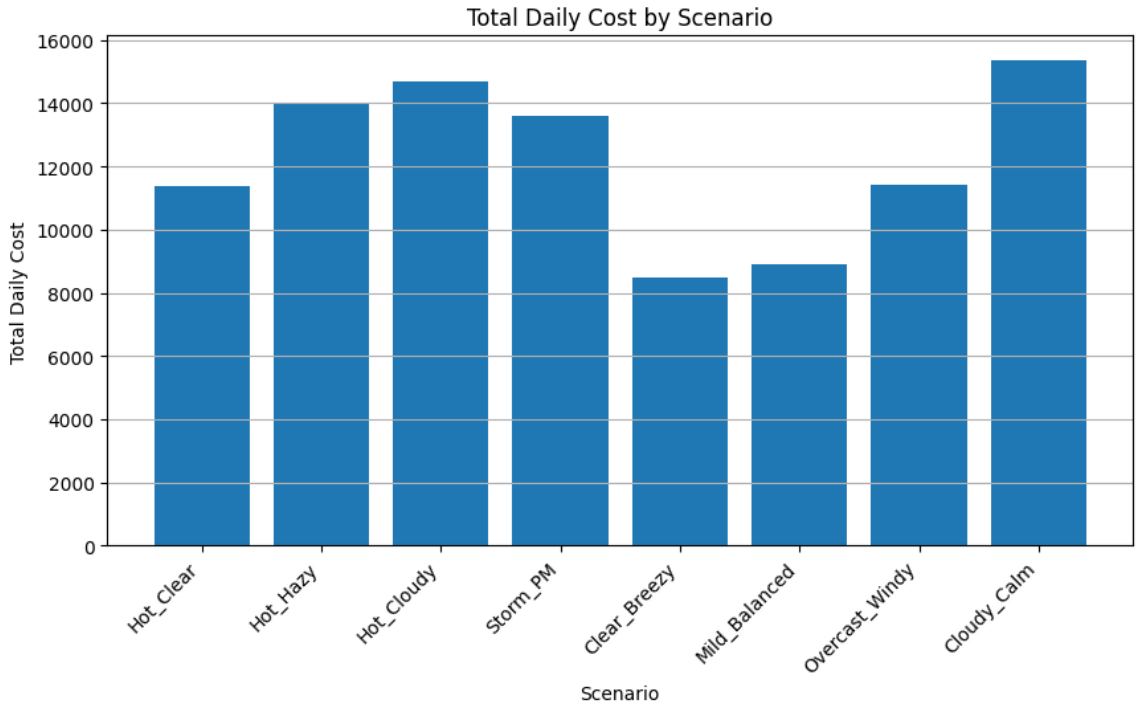

Figure C. Total daily system cost by scenario from the Python model

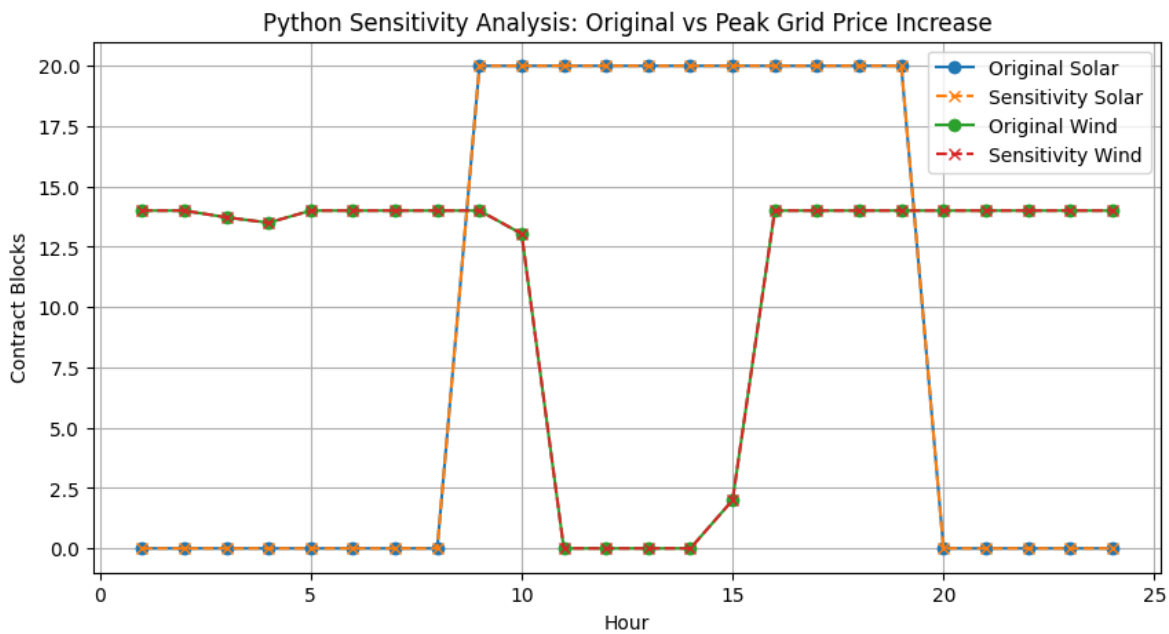

Figure D - Python Sensitivity Comparison.